

Benthic Habitat Segmentation

CISC684: Introduction to Machine Learning

1. Introduction

This project develops a deep learning model for benthic habitat segmentation using WorldView-3 multispectral satellite imagery from Guam. The task is to classify each image pixel into one of seven habitat classes: Background, Seagrass, Coral, Macroalgae, Sand, Land, and Ocean. The dataset contains 8-band GeoTIFF images from Agana Bay and Manell-Geus, with corresponding expert-annotated masks for training and evaluation. The main challenges are variable image sizes, strong class imbalance, and spectral ambiguity among underwater habitat classes such as seagrass, coral, macroalgae, and sand. The assignment requires preprocessing, model development for 8-channel input, train/validation analysis, testing on unseen Agana and Manell-Geus images and error visualization.

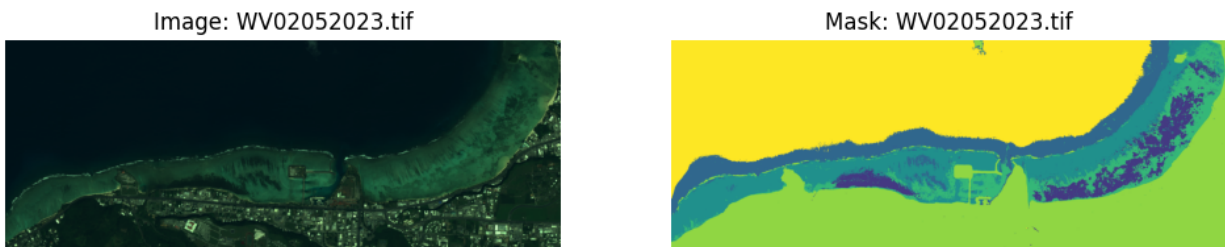


Figure 1. Example labeled training image. The left panel shows an 8-band WorldView-3 image visualized using selected RGB-style bands, and the right panel shows the corresponding ground-truth habitat mask. This example shows the pixel-level segmentation target used for training.

2. Methodology

2.1 Data Preprocessing

The dataset contains 9 labeled training images and 21 test images, including 15 Agana Bay images and 6 Manell-Geus images. Since the original images have different spatial resolutions, a patch-based strategy was used instead of resizing the full images. During training, each image was sampled into fixed-size 256×256 patches. This allowed the model to process all images using a consistent input size while preserving local spatial detail.

Each input image contains 8 spectral bands: Coastal, Blue, Green, Yellow, Red, Red Edge, NIR1, and NIR2. All bands were converted to floating-point values and normalized using the mean and standard deviation computed from the labeled training images. The band-wise normalization was:

$$x'_b = x_b - \mu_b / \sigma_b + \epsilon$$

where x_b is a pixel value from band b , and μ_b and σ_b are the training-set mean and standard deviation for that band.

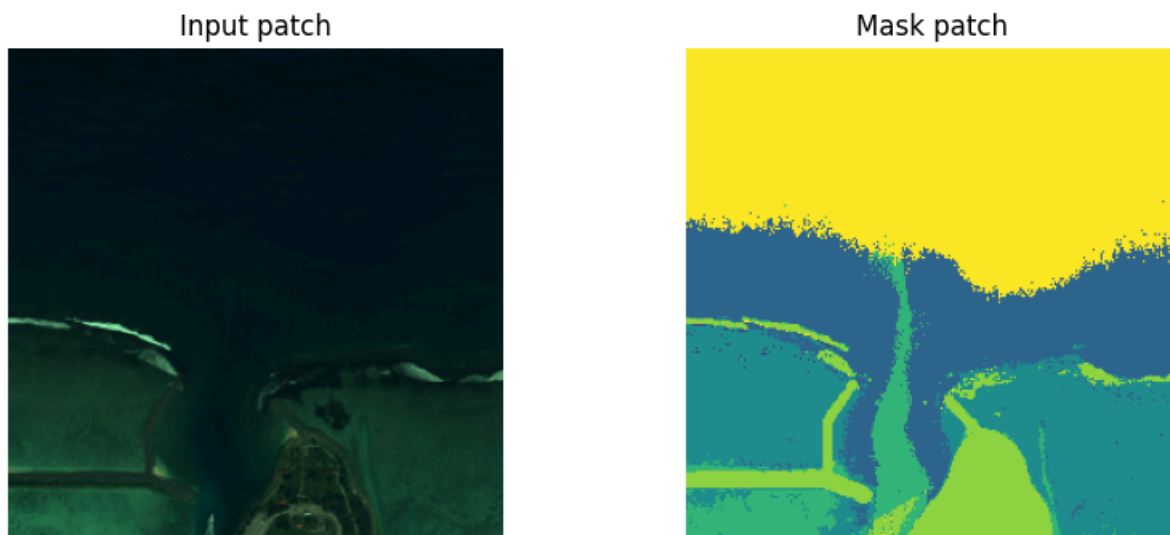


Figure 2. Patch-based training sample. Variable-sized satellite images were divided into fixed 256×256 patches. The left panel shows an input patch and the right panel shows the corresponding mask patch. Patch-based training avoids resizing full images and preserves local spatial detail.

The dataset is highly imbalanced. Land and Ocean dominate the pixel distribution, while ecologically important classes such as Seagrass and Coral are much less frequent. To address this, two strategies were used. First, weighted cross-entropy loss assigned larger weights to rare classes. Second, balanced patch sampling was used during training: 70% of sampled patches were centered around rare habitat classes, including Seagrass, Coral, Macroalgae, and Sand.

Class ID	Class Name	Pixel Count	Pixel Percentage	Loss Weight
0	Background	608675	3.50%	1.989
1	Seagrass	532751	3.06%	2.272
2	Coral	1331689	7.65%	0.909
3	Macroalgae	2072909	11.91%	0.584
4	Sand	1496590	8.60%	0.809
5	Land	6596518	37.89%	0.184
6	Ocean	4771838	27.41%	0.254

Table 1. Class distribution and weighted loss values used to address class imbalance. Rare classes such as Seagrass and Background receive larger loss weights, while dominant classes such as Land and Ocean receive smaller weights.

2.2 Model Architecture

An encoder-decoder U-Net architecture was used for semantic segmentation. U-Net was selected because it is well-suited for pixel-level prediction and preserves fine spatial information using skip connections between encoder and decoder stages. The model was modified to accept 8 input channels instead of the standard 3 RGB channels. The final output layer produces 7 channels, corresponding to the 7 habitat classes.

The encoder consists of repeated double-convolution blocks with batch normalization, ReLU activation, dropout, and max pooling. The decoder upsamples feature maps using transposed convolutions and concatenates them with encoder features through skip connections. These skip connections help recover boundary details and improve segmentation of small habitat regions.

Parameter	Value
Model	Small U-Net
Input channels	8
Output classes	7
Patch size	(256 x 256)
Batch size	16
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Epochs	25
Loss function	0.7 Weighted Cross-Entropy + 0.3 Dice Loss
Augmentation	Horizontal and vertical flips
GPU	NVIDIA A100-SXM4-40GB

Table 2. Training configuration for the proposed 8-channel U-Net model.

2.3 Loss Function and Evaluation metrics

The training objective combined weighted cross-entropy loss and multiclass Dice loss:

$$L=0.7L_{CE}+0.3L_{Dice}$$

Weighted cross-entropy helps compensate for class imbalance, while Dice loss improves overlap between predicted and ground-truth masks. This combination is more appropriate than standard cross-entropy alone because the dataset contains rare but important classes.

The model was evaluated using pixel accuracy and mean Intersection over Union. Pixel accuracy measures the percentage of correctly classified pixels:

$$\text{Pixel Accuracy} = \text{Correct Pixels} / \text{Total Pixels}$$

For each class, IoU is computed as:

$$\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$$

Mean IoU is the average IoU over all classes. Since the dataset is imbalanced, mIoU is more informative than pixel accuracy because it penalizes poor performance on rare classes.

3. Experimental Results

The model was trained on 7 labeled images and validated on 2 labeled images, including one smaller Agana-style image and one larger Manell-Geus-style image. The best checkpoint was selected using validation mIoU rather than the final epoch. This was done because validation loss fluctuated due to the small dataset size and patch-based validation, while mIoU directly measures segmentation quality.

The best validation mIoU was approximately 68.5%. The final saved checkpoint was then used for sliding-window inference on the 21 unseen test images. During testing, overlapping 256×256 patches were predicted with stride 128, and overlapping probabilities were averaged to reconstruct the full-size segmentation mask.

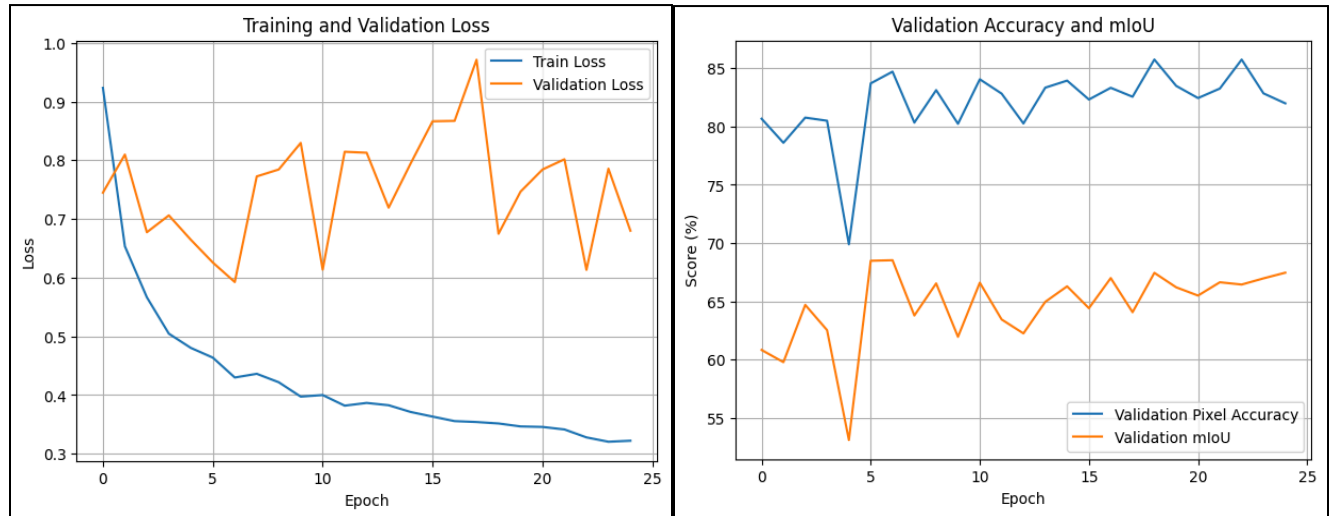


Figure 3. Training and validation behavior over 25 epochs. Training loss decreases steadily, while validation loss fluctuates due to the small dataset and patch-based sampling. Validation mIoU remains relatively stable after early epochs, so the best validation-mIoU checkpoint was selected for testing.

The overall test pixel accuracy was 79.33%, and the overall test mIoU was 63.50%. Agana Bay achieved slightly higher mIoU than Manell-Geus, while Manell-Geus achieved higher pixel accuracy. This suggests that the model performs well on dominant regions such as Land and Background, but still struggles with some underwater habitat boundaries and rare classes.

Test Region	Pixel Accuracy	mIoU
Agana Bay	77.69%	64.56%
Manell-Geus	83.43%	60.86%
Overall	79.33%	63.50%

Table 3. Test performance on unseen Agana Bay and Manell-Geus images.

The complete per-image metrics and per-class IoU values are provided in test_metrics.csv in the submitted results folder.

4. Error Analysis

The qualitative results show that the model captures the major spatial structure of the scenes, including Land, Ocean, and broad reef/habitat regions. However, errors are concentrated near class boundaries and in mixed underwater regions. In particular, confusion occurs between Coral, Seagrass, Macroalgae, and Sand because these classes can have similar spectral and spatial patterns in shallow-water satellite imagery.

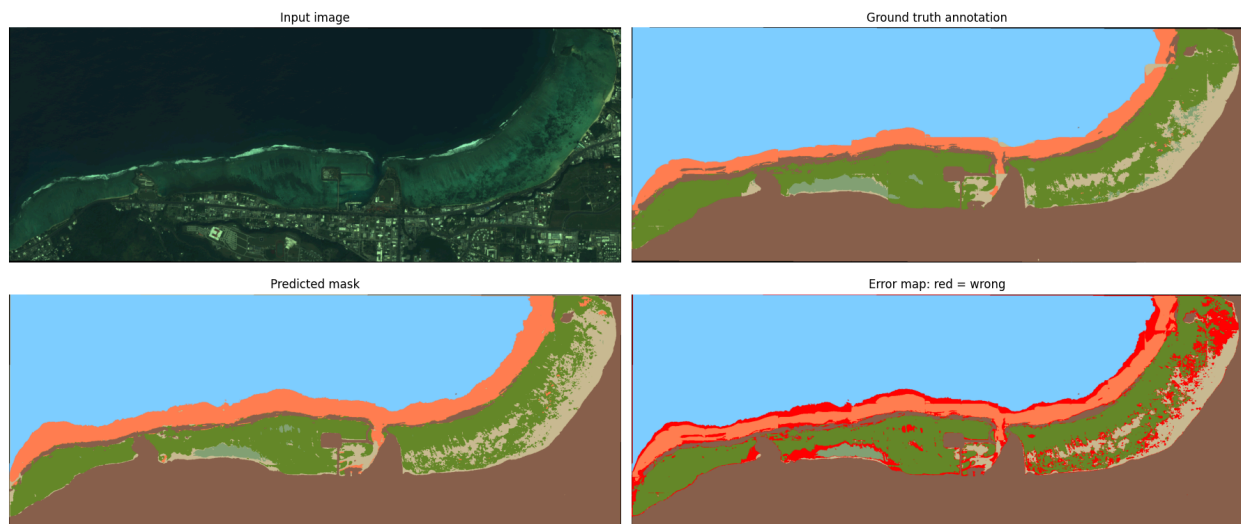


Figure 4. Example segmentation result for an Agana Bay test image. The model correctly identifies large Land, Ocean, and reef regions, but the red error map shows mistakes near habitat boundaries and mixed benthic regions.

For the shown Agana Bay example, the model achieved 93.19% pixel accuracy and 73.58% mIoU. The strongest classes in this example were Land and Ocean, with IoU values of 96.53% and 94.79%, respectively. Macroalgae was also segmented well with an IoU of 79.58%. Seagrass was more difficult, with an IoU of 33.76%, showing that rare habitat classes remain challenging.

Across the full test set, some images had very low Ocean IoU, especially several Agana examples. This indicates that the model sometimes misclassified water regions when their spectral appearance differed from the training distribution. The Manell-Geus 2024 image also

had low mIoU, suggesting a possible domain shift due to different acquisition conditions, water depth, or turbidity.

Overall, the main error sources were:

- Confusion between visually similar benthic classes, especially Seagrass, Coral, Macroalgae, and Sand.
- Boundary errors near land-water transitions and reef edges.
- Reduced generalization on images with different spectral conditions or acquisition dates.
- Lower performance on rare classes compared with dominant Land and Ocean regions.

5. Conclusion

This project implemented an 8-channel U-Net model for benthic habitat segmentation using WorldView-3 multispectral imagery. The preprocessing pipeline handled variable image sizes using patch-based training and sliding-window inference. Band-wise normalization, weighted cross-entropy, Dice loss, and rare-class patch sampling were used to address the challenges of multispectral input and class imbalance.

The final model achieved 79.33% overall pixel accuracy and 63.50% overall mIoU on the unseen test set. The results show that the model can segment large habitat structures effectively, especially Land, Ocean, and broad reef regions. However, rare and spectrally similar underwater classes remain more difficult. Future improvements could include more labeled training data, stronger spectral-spatial augmentation, larger U-Net backbones, or attention-based segmentation models to improve rare-class accuracy and boundary precision.